Measurement Systems

An Open-Source Summary

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Contents

> **Note**
>
> This summary will only be based on the slides and lectures. No figures or copy from the cursus book is permitted due to copyrights.
>
> Nonetheless, it is recommended to also go through the cursus book as it may go more in depth than this summary

1 Introduction

1.1 Why is measuring important ?

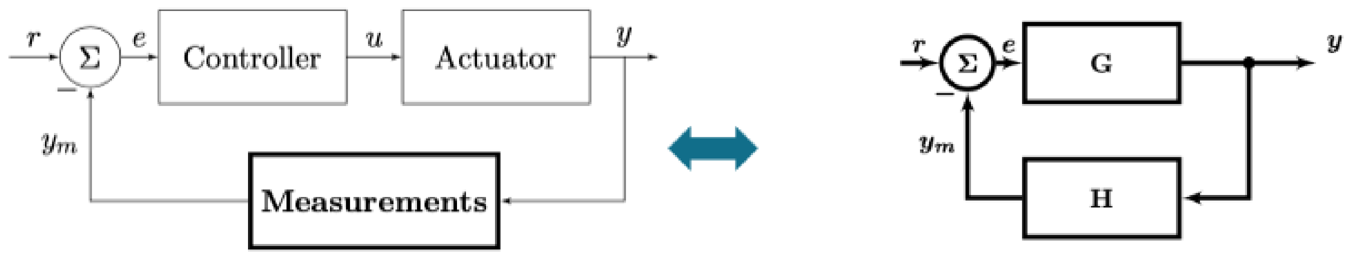

Figure 1: Typical system

In a feedback system, we want some input to transform in a specific output. Later, we want to measure this output to adapt the control and see how effective this is.

Using the standard G and H notation we can find that:

$$y = \frac{G}{1+GH}r \qquad e = \frac{1}{1+GH}r \tag{1}$$

$$y = \lim_{G\to\infty} \frac{G}{1+GH}r \approx \frac{1}{H}r \tag{2}$$

1.2 General principles

> Building a measurement system is transforming energy/information from one domain to another.

We first start by building the physical knowledge (build physical quantity model), mathematical reasoning (find how accurate we can make the model to reflect the system) and finally we must take noise and other limits into consideration.

In a system, at a high-level, there is 2 forms of variables:

1. **Across Variables:** describe the *state*, in **parallel** with the terminal. Effort to change the state
2. **Through Variables:** describe the *flow*, in **series** with the terminal.

Then combining those two can give us some fundamental quantities: - $A \cdot T$: **power** generated or dissipated in the element - A/T: **impedance** of the system. Describes how an element transforms an A into a T

1.3 Modelling a system : the physical model

We usually want to map the IO behavior based on physical knowledge. Each model tries to represent reality but will always introduce errors or not take into account higher order phenomena.

For example, we can model a pressure change using a diaphragm to a capacitance change. If we take a simple spring model we don't take second order term into account which introduces errors.

Table 1: Strength and weaknesses of physical approach

Strength	Weakness
Good insight	Error due to approximations
System optimization is possible	Derivation of this formula is not always straightforward/possible
Calibration techniques can be easily derived	Errors due to tolerances in fabrication

1.4 Correlation

A typical operation used in system is **correlation**, it is used to find the "likelihood" between two signals. It helps us extract features:

$$R_{fg}(t) = \int_{-\infty}^{\infty} f(t)g(t-\tau)d\tau$$

Knowing Fourrier's theory, we can represent any signal as a combination of sin and cos. Instead of using $g(t-\tau)$, this gives us:

$$
\begin{aligned}
F(f) &= F_{cosine}(f) + F_{sine}(f) && (3)\\
&= \int_{-\infty}^{\infty} (cos(2\pi ft) + isin(2\phi ft))f(t)dt && (4)
\end{aligned}
$$

1.4.1 Freshen up of Laplace transform

The Fourier transform can be written as:

$$\mathcal{F}f(t) = \int_{-\infty}^{\infty} e^{i2\pi ft} f(t)dt$$

The Laplace transform goes one step further by using the Euler notation $e^{ix} = cos(x) + isin(x)$ and extend the Fourier transform with complex numbers:

$$\mathcal{L}f(s) = \int_{0}^{\infty} e^{-st} f(t)dt \qquad s = \sigma + i\omega t$$

It can be seen that:

$$\mathcal{F}(f(t)) = \mathcal{L}(f(t)) \iff f(t) = 0 \text{ if } t < 0$$

Laplace transform is an easy tool to solve differential equation and work with derivative:

$$\mathcal{L}\left(\frac{df(t)}{dt}\right) = s\mathcal{L}(f(t)) + f(0)$$

1.5 The Measurement Chain

> A measurement system converts a state variable in a measured value

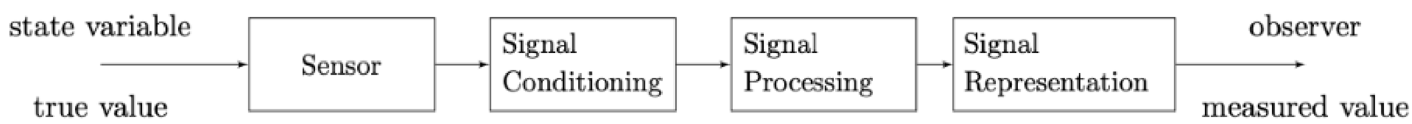

Figure 2: 4 major components

The idea behind a *signal conditioning* is to reduce error and filter out the signal. This should also convert it into a digital signal that can be easily processed by a DSP block. DSP block is used to work with the input and realize complex function while the signal representation can be a screen, plot, …

1.5.1 4 major components

1.5.1.1 Sensor Each sensor is made of 2 sensors to be exact.

1. A primary that transform a physical quantity to an "*easier to measure*" quantity
2. A secondary that converts this quantity into an electrical signal

1.5.1.2 Signal conditioning This is usually an ADC block. Such block are limited by the current technologies and we call that a **technological constant** C_T. There is no free lunch and everything is a tradeoff leading to a basic formula from information theory:

$$\frac{Speed(Accuracy)^2}{Power} = C_T$$

An ADC needs an input **and** a reference voltage. I outputs $\frac{V_{in}}{V_{ref}} = \frac{N}{2^{n-1}}$. The flash converter using a resistive ladder is a typical (but bad way) to convert into digital. We also need a clock and comparators to convert.

Accuracy is important and the Effective Number of Bits or ENOB is a crucial metric for any system. Due to quantization error and the SNR required for 1 valid bit, we can find such formula[1] :

$$ENOB = \frac{SNDR - 1.76dB}{6.02dB}$$

So in summary, building a physical models is a good step in the right direction to be able to better filter and analyze measurements.

2 Characterization of measurements

2.1 Measurement Errors

In any system, there will be error which will reduce its accuracy. The error is added after each operation in our model but it is important to understand how it **propagates** and its **origin**.

Table 2: Comparing the two type of errors

Characteristics	Absolute error	Relative error
Math	$e = x_m - x$	$e_r = \frac{e}{x} = \frac{x_m - x}{x}$
Unit	Same unit as variable	Unitless - ppm
Estimation	Over-estimation, $e \propto G$	Under-estimation, $e_r! \propto G$

$$x_m = S(s)G(s)x + G(s)\bar{e_s} + \bar{e_c} \quad \text{Output measurement} \quad \bar{e_m} = G(s)\bar{e_s} + \bar{e_c} \quad \text{Error of this measurement} \tag{5}$$

$$y = \frac{x_m}{S(0)G(0)} \quad \text{Transfer function} \quad e_i\bar{n} = \frac{\bar{e_s}}{S(s)} + \frac{e_c}{S(s)G(s)} \quad \text{Input referred noise} \tag{6}$$

[1]Find more information about how we can get such formula in the DAMSIC course, ***link to a summary of the course***. Check the section 4.1.2

Table 3: Type and difference of errors

Type	Effect	Example
Interfering	Additive to the signal	Magnetic interferences
Modifying	Modifies transfer function	Pressure sensor changes over deformation

By understanding the nature of an error, we can reduce it or even make it disappear. Example, EM interference can disrupt ESG signals. Solution, add an extra probe (right leg) to also capture this interference and connect it to the V_{ref}. Now, V_{in} and V_{ref} in the ADC will have the same error which will cancel out.

2.2 Error Propagation

We must understand how an error propagates in a system. We can build a simple model like:

$$x_m = ax + by$$

Where x is the true value that will be measured x_m, but also possible interferance such as y. The error is:

$$e_m = a\Delta x + b\Delta y$$

We can also conduct some statistical analysis:

$$\begin{aligned} Var(x_m) &= \mathbb{E}\left[(ax+by) - (\mathbb{E}(ax+by))^2\right] && (7)\\ &= ... && (8)\\ \sigma_m^2 &= a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\sigma_{xy} && (9) \end{aligned}$$

The last term is often forgotten because a lot of person assume $\perp$ between x and y which is not always the case. The covariance is null only if uncorrelated. So typically, if the errors are due to the same source, their covariance is most likely correlated.

2.2.1 Non-linear functions

For more advance transfer function that are non-linear, we can rewrite it using the Taylor series:

$$f(x,y) = f(x_1,y_1) + \frac{\partial f}{\partial x}(x-x_1) + \frac{\partial f}{\partial y}(y-y_1) + ...$$

Assuming a small error we can simplify our function and say that

$$\sigma_m^2 \approx \frac{\partial f}{\partial x}^2\sigma_x^2 + \frac{\partial f}{\partial y}^2\sigma_y^2 + 2\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}\sigma_{xy}$$

2.2.2 Static Characteristics

- **Range & Span**: range is the min and max while the span is the delta between the two
  - Those values should reflect where the system is still meeting its specifications

2.2.2.1 Linearizing the reading Knowing the range, we could force a simple linearization using:

$$O - O_{min} = \frac{O_{max} - O_{min}}{I_{max} - I_{min}}(I - I_{min})$$

But this is far from ideal, a better solution is the least square solution:

$$K = \frac{Cov(X_1, Y_1)}{Var(X_1)} \tag{10}$$

$$= \frac{X_1 Y_1 - n\overline{X_1 Y_1}}{X_1 Y_1 - n\overline{X_1}^2} \tag{11}$$

Table 4: Quick overview of the metrics

Metric	Description	Issue
Max error	Maximum error from ideal and model	Doesn't give a real insight
DNL	Difference between width of the step and ideal LSB step. <1LSB or problematic	Watch out for sign inversion
INL	Maximum deviation (max(DNL))	Less insight
THD	Each non-linearity causes distortion, highlight the distortion nature	Doesn't give information about the non-linearity characteristic

It is important to understand the error of the non-linearity. Because compensating for it is costly as we will use a polynomial to cancel it. The order of the polynomial will require $n + 1$ calibration measurements.

2.2.2.2 Resolution

> **Definition**
>
> Resolution is the smallest discrete step a system can take
>
> $$resolution = \frac{\Delta I_R}{I_{max} - I_{min}} \cdot 100\%$$

The resolution isn't the accuracy as we could add dummy 0 which won't improve the accuracy but the resolution.

2.2.2.3 Hysteresis It is a truly physical phenomena that depicts how when changing the input and reducing it will not lead to the same output path due to various reasons.

2.2.2.4 Dynamical errors When measuring something, we often think we are in the steady state which is not really the case. Most of the time we will wait x amount of time until it is "*good enough*".

This is all linked with the **memory elements** that is present in mechanical, electrical, $\dots$ systems. The more **independent** memory elements we have the higher is the order of the system. We talk here about settling time which models the *exponential* behavior of such event;

$$\varepsilon = \Delta I expt \left(-\frac{t}{\tau}\right)$$

Of course, Taylor series can be used for small amplitude: $TF(s) = \frac{\partial O}{\partial I}|_{I=I^*} 1/(1+s\tau)$

2.3 Measurement Characteristics

3 License

3.1 Contributions

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